

Hail Song

PHD STUDENT · KAIST UVR LAB

Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea

✉ hail96@kaist.ac.kr | 🏠 hailson.github.io | 🔗 linkedin.com/in/hailson

Education

Korea Advanced Institute of Science and Technology

PH.D. STUDENT, GSCT (MINOR: GRADUATE SCHOOL OF METAVERSE)

- Advisor: Prof. Woontack Woo

Daejeon, Republic of Korea

Mar 2024 - present

Korea Advanced Institute of Science and Technology

M.S., GSCT (MINOR: GRADUATE SCHOOL OF METAVERSE)

- Advisor: Prof. Woontack Woo
- Thesis : Real-time Cumulative SMPL-based Avatar Body Generation System

Daejeon, Republic of Korea

Mar 2022 - Feb 2024

Yonsei University

B.S., MECHANICAL ENGINEERING

Seoul, Republic of Korea

Mar 2015 - Feb 2021

Experiences

- Aug 2024 - **Visiting Scholar at Carnegie Mellon University**, Sponsored by Institute of Information & Communications Technology Planning & Evaluation, IITP.
- Feb 2025 - **Visiting Scholar at TUM (Technical University of Munich)**, TUM School of Computation, Information and Technology, CAMP
- Jul 2023 - **Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
- Aug 2023 - **Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
- Mar 2022 - **Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
- present - **Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
- May 2020 - **Computer Vision Researcher (Full-time)**, InBody Co., Ltd.
- Feb 2022 - **Computer Vision Researcher (Full-time)**, InBody Co., Ltd.
- Oct 2016 - **Mandatory Military Service**, ROK Army sergeant, honorable discharge
- Jul 2018 - **Mandatory Military Service**, ROK Army sergeant, honorable discharge

Research Interests

HUMAN AVATARS WITH NEURAL RENDERING

My research focuses on reconstructing high-fidelity 3D human avatars from minimal and accessible inputs using neural rendering. My work spans single-image head avatar generation, full-body avatar reconstruction from monocular RGB-D video [ISMAR 2023], and sketch-driven 3D face modeling [CVPRW 2025]. I aim to make avatar creation more scalable, practical, and accessible for everyday users.

REAL-TIME AVATAR SYSTEMS FOR XR

I am particularly interested in deploying high-fidelity avatars in immersive XR environments. My work addresses systems-level challenges such as real-time neural rendering, efficient stereo display, expression control from HMD sensors, and motion estimation from sparse egocentric input. I aim to bridge avatar reconstruction and interactive XR deployment, enabling real-time, responsive avatar-mediated communication on practical head-mounted devices for both full-body [TVCG 2026] and head [TVCG 2026] avatars based on 3D Gaussian Splatting. More recently, I have been exploring robust full-body motion generation under sparse tracking conditions.

Publications

FULL PAPERS (JOURNAL AND CONFERENCE PAPERS)

VRGaussianAvatar: Integrating 3D Gaussian Avatars into VR

Hail Song*, Boram Yoon, Seokhwan Yang, Seoyoung Kang, Hyunwoo Kim, Hannes Metzmacher, Woontack Woo
IEEE Transactions on Visualization and Computer Graphics (**TVCG 2026**)

OFERA: Blendshape-driven 3D Gaussian Control for Occluded Facial Expression to Realistic Avatars in VR

Seokhwan Yang, Boram Yoon, Seoyoung Kang, **Hail Song**, Woontack Woo
IEEE Transactions on Visualization and Computer Graphics (**TVCG 2026**)

Streamlined Facial Data Collection based on Utterance and Emotional Data for Human-to-Avatar Reconstruction

Seoyoung Kang, Seokhwan Yang, **Hail Song**, Boram Yoon, Jaehoon Kim, Kangsoo Kim, Woontack Woo
IEEE Transactions on Visualization and Computer Graphics (**TVCG 2026**)

SceneLinker: Compositional 3D Scene Generation via Semantic Scene Graph from RGB Sequences

Seok-Young Kim, Donghyeon Kim, Woojin Cho, **Hail Song**, Seoyoung Kang, Woontack Woo
IEEE Transactions on Visualization and Computer Graphics (**TVCG 2026**)

The Influence of Emotion-based Prioritized Facial Expressions on Social Presence in Avatar-mediated Remote Communication

Seoyoung Kang, **Hail Song**, Boram Yoon, Kangsoo Kim, Woontack Woo
2024 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2024**)

RC-SMPL: Real-time Cumulative SMPL-based Avatar Body Generation

Hail Song, Boram Yoon, Woojin Cho, Woontack Woo
2023 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2023**)

SHORT PAPERS (WORKSHOPS AND DEMOS)

HMD-only Controllable 3D Gaussian Avatars in VR: Face and Full-body Demonstration

Seokhwan Yang, **Hail Song**, Woontack Woo
2026 IEEE Virtual Reality Workshop (Research Demo) (**VRW 2026**) [Best Research Demo Award Nominee](#)

VR Zen Garden: Designing a Virtual Environment for Stress Relief

Hail Song, Jinseok Hong, Seonji Kim, Kyung Taek Oh, Woontack Woo, Sungyoung Kim
2026 IEEE Virtual Reality Workshop (Research Demo) (**VRW 2026**)

Fast Texture Transfer for XR Avatars via Barycentric UV Conversion

Hail Song, Seokhwan Yang, Woontack Woo
2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (**ISMAR Adjunct 2025**)

S3D: Sketch-Driven 3D Model Generation

Hail Song*, Wonsik Shin*, Naeun Lee, Soomin Chung, Nojun Kwak, Woontack Woo
GMCV Workshop at CVPR 2025 (**CVPRW 2025**)

Toward Realistic 3D Avatar Generation with Dynamic 3D Gaussian Splatting for AR/VR Communication

Hail Song
2024 IEEE Virtual Reality Workshop (Doctoral Consortium) (**VRW 2024**) [Best DC Paper Award](#)

DOMESTIC CONFERENCE PAPERS

Meta-Rodin: A Multi-Sensory Immersive Sculpting System to Cultivate a Memorable Experience in a Meta-Museum

Sihyun Jeong, Coulon Jina Jung, **Hail Song**, Woontack Woo
Korea Computer Congress 2024 (**KCC 2024**) [Outstanding Presented Paper Award](#)

Quantitative Optimal Learning Conditions for Gaussian Splatting Based on Input Quantity and Scene Dynamics

Hyungwoo Jin, Eunhee Jung, **Hail Song**, Woontack Woo
Korea Software Congress 2023 (**KSC 2023**)

Real-Time Music Tracking System in Wearable AR Display for Music Performance

Jiyun Park, **Hail Song**, Seungmin Lee, Juhan Nam, Woontack Woo
2023 The HCI Society of Korea (**HCI Korea 2023**)

Awards, Fellowships, & Grants

2026	Certificate of Excellence (Edu4.0 Q TA) , KAIST Center for Excellence in Learning & Teaching	
2024	IEEE VR 2024 Best DC Paper Award , IEEE VR 2024	
2024	KCC Outstanding Presented Paper Award , KCC 2024	
2020	Department Honors , Yonsei University	
2015	National Science & Technology Scholarship , KOSAF (Korea Student Aid Foundation)	\$ 26,000

Teaching Experience

2026	GCT555 3D Interaction Design , Spring 2026, Teaching Assistant	KAIST, Daejeon
2025	GCT600/MV600 Augmented Reality Project , Fall 2025, Teaching Assistant	KAIST, Daejeon
2025	CTP445 Augmented Reality , Summer 2025, Teaching Assistant	KAIST, Daejeon
2019	Teaching Assistant (Coding) , Samsung Dream Class, supported by Samsung Co., Ltd.,	Incheon

Academic Activities

INVITED TALKS

2023 Korea Software Congress (KSC), Top-Conference Session

REVIEWER

IEEE ISMAR 2024, 2025 / IEEE VR 2026 / Korea Computer Congress (KCC) 2024, 2025